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Foam formation and control in bioreactors

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Abstract

Foam formation during the course of a bioprocess remains a major technological challenge to be resolved and needing further investigation. The foaming tendency of the cultivation media used in bioreactor induces a lot of direct, i.e. microbial cells stripping, contamination, as well as indirect adverse effects, i.e. modification of the properties of the medium subsequently to the addition of chemical antifoam leading to toxic effect at the level of the microbial metabolism and fouling of the downstream processing equipments. In this work , the mechanisms leading to foam formation will be described at the molecular level (adsorption of surface active molecule at the gas-liquid interface), but also at the process level with a special attention given at the level of the compatibility of the intensification of the gas-liquid operation in bioreactors in front of the potential foam associated problems. In front of these considerations, both the chemical and mechanical antifoam techniques have evolved. A lot of new antifoam formulations, as well as original combinations of chemical antifoams have been especially designed in order to meet the bioprocesses specific requirements. Original mechanical techniques to prevent the foam formation in bioreactors have been elaborated

from the combined knowledge involving the fluid dynamics of the gas-liquid dispersion and the interfacial processes (e.g. the "Stirring As Foam Disruption" concept").

As outlined by Prins and van't Riet in 1987 [1], foam in bioprocess operations has received little attention compared with other engineering problem such as mixing and mass transfer. Since this, the problem of foam formation in fermentation processes is not yet well controlled as reported in a recent review [2]. Most of the progresses have been made at the level of the knowledge of the physico-chemical mechanisms and only a few researches have been performed on the tools needed to control foam during a bioprocesses. These knowledges have notably led to the conception of chemical antifoams. However, the use of such chemicals in bioprocesses must be carefully monitored, considering the microorganism sensitivity to extracellular changes. A second source of important data comes from the investigation of the hydrodynamic effect leading to the appearance of foam in bioreactors.

1. A first overview of the mechanisms of foam formation in bioreactors

In bioreactors, the intensification of the aeration process and the protein content of the broth are at the basis of foam formation. Indeed, in aerobic culture, oxygen is provided by the injection of compressed air at the lower part of the reactor. The oxygen solubility being very low in aqueous media (about 7-8 mg/l in the fermentation conditions) and the oxygen requirement by the microorganisms being very high, the oxygen transfer from air bubbles to the liquid phase must be strongly optimized. This is achieved in practice by increasing the air flow rate through the reactor and, in the case of stirred bioreactors, by increasing the stirrer

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3 speed. In all cases, the oxygen transfer process is optimized by increasing the gas-liquid
4 dispersion inside the vessel. The rising bubbles (being as fine as possible) cross the surface of
5 the liquid inside the reactor and in the presence of surface active molecules induces the
6 formation of foam. In general, the surface active molecules found in fermentation media are
7 proteins at the basis of most of the industrial complex media or being released by cell
8 autolysis during the course of the process [3]. Other compounds exhibiting a surface active
9 behavior can also be secreted by microorganisms, e.g. *Escherichia coli* is known to release
10 lipopolysaccharides (LPS) when exposed to starvation stress during a culture [4] and *Bacillus*
11 *subtilis* is able to synthesize several lipopeptides [5]. A wide variety of compounds excreted
12 during the culture of *Penicillium herqueii* have been characterized. These include pigments
13 and non identified lipophilic surfactants coming from cell lysis and being accumulated in the
14 foam [6]. However, proteins are the most frequent surface active agent encountered in
15 bioprocesses. The proteins are adsorbed at the gas-liquid interfaces, and according to their
16 surface active properties (proteins are constituted by chains of amino acids exhibiting each
17 hydrophobic or hydrophilic properties) are able to promote foam at very low concentrations
18 (as low as 1 mg/l [1]). In addition, it has been shown that the sterilization procedure of a
19 protein solution increases significantly the foaming capacity of the medium [3]. Considering
20 the isoelectric point of the protein, i.e. the pH for which the global charge of the protein is
21 null, the pH of the medium plays an important role on the foam formation. However, the pH
22 is also an important parameter which affects microbial growth and is usually controlled during
23 the culture and cannot be modified in practice to limit foam. In some specific cases, surface
24 active molecules, named biosurfactants, can be released by the microorganism secondary
25 metabolism [7, 8], leading to an increase of the process difficulties promoted by the foaming
26 media. To illustrate this problem, some researches involving a *Bacillus subtilis* strain
27 overproducing a surfactin (a surface active molecule belonging to the lipopeptide class) have
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necessitated a special reactor configuration with oxygen provided by surface aeration in order to avoid the problem associated with air injection [9]. Foam, owing to the amount of precautions to be taken to prevent its formation, is a big challenge for process engineer. The problem generated by foaming medium are multiple [10]: decrease of the working volume of the reactor, removal of microbial cells and metabolites, contamination, decrease of the oxygen transfer rate subsequently to the addition of antifoam, toxic effects on cell physiology,...

Two spatial scales are involved in the understanding of the underlying mechanisms associated with the foam formation in bioreactor. First, the microscopic scale involving the molecular arrangement at the gas-liquid interfaces. Second, the macroscopic scale involving the gas-liquid flows at the level of the bioreactor.

2. Description of the foam at the microscopic level : molecular mechanisms at the interfaces

An extensive amount of informations about the physical and molecular mechanism at the basis of the foam formation are available in the literature. In this section, only the basic mechanisms involving the molecular mechanisms at the basis of foam formation will be presented. The molecules comprising a hydrophobic and a hydrophilic part are termed as surface active and are preferentially located at the interfaces. The surface tension, i.e. the interfacial free energy par unit area, represents the force of cohesion that maintains the molecules in each phase. A surface active molecule is able to reduce significantly the surface tension after its adsorption at the interface. One of the major macroscopic effect resulting from this adsorption is the stabilization of the dispersion of one phase to the another. In aqueous solution, when the air-water interface is saturated, the adsorption is stopped and the surface excess of surface active molecules is arranged in micelles into the solution. This point

corresponds to the critical micellar concentration (CMC), i.e. the surfactant concentration leading to the saturation of the interfacial film. Above the CMC, the surfactant concentration has no influence on the surface tension (Figure 1). The formation of micelles is also a mechanism by which the contact between the water and the hydrophobic parts of molecules is minimized, leading to a reduction of the free energy of the system. However, this is not the magnitude of the surface tension drop that influences the formation of foam. Indeed, there are surface active molecules reducing greatly the surface tension but leading to unstable foams. The concept of stability of the foam is thus important and appears to be strongly correlated to the rate of surface tension variation $d\sigma$ [11]. An important parameter controlling the stability of the interface against the mechanical actions is the elasticity force [12, 13]. The effect of surface dilational elasticity has first been described by Gibbs by the following equation:

$$E = \frac{d\sigma}{d\ln A} \quad (1)$$

With being σ the surface tension and $d\ln A$ the relative change in surface area.

Equation 1 is very relevant since the flow properties of liquid/air interfaces are strongly influenced by elastic forces.

It is important to point out the molecular differences between foam produced with low molecular weight surfactant and proteins. Indeed, proteins differ from low molecular weight surface active molecules by several aspects : it is very difficult to desorb a protein from an interface except with a low molecular weight surfactant, protein can change their conformation at the interface giving some elasticity to the foam and consequently increasing the foam stability, more than one layer of protein can adsorb at the interface [1]. In summary, the interfacial effect induced by the protein adsorption is strongly linked with its conformation at this interface (see figure 2 for a comparison of the conformation of a low molecular weight surfactant and a protein at an interface). Under dynamic conditions, such as those encountered in bioreactor where bubbles continually flow through the media and burst at the free surface,

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3 proteins can change their conformation in order to readjust the surface tension and the
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5 equilibrium. This is a good example of elasticity leading to foam with a high stability which
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7 can be quantified by the Gibbs dilational modulus (equation 1). Indeed, in the case of protein
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9 stabilized foam, this modulus is very high, which means that surface tension gradient can
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11 persists for a long time in a surface to which protein is adsorbed. However, the adsorption of
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13 surfactants at the interface is not the only conditions required to lead to the formation of foam.
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15 The gas-liquid mixture depending on the intensification of the aeration process in the reactor
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17 is a very important factor, the surfactant being only required in order to stabilize the bursting
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19 bubbles at the level of the free surface. These hydrodynamic phenomena will be described in
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21 the next section.
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29 **3. Description of the foam at the macroscopic level : gas-liquid dispersion in the** 30 **reactor** 31 32 33 34 35

36 In the previous section, it has been shown that the adsorption of surfactant molecules at the
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38 gas-liquid interface is required in order to produce foam. However, this is not the only
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40 condition. Indeed, the gas-liquid mixture inside the bioreactor is also important, the surfactant
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42 being only required to stabilize the bursting bubbles at the liquid free surface. Several authors
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44 have pointed out the fact that the turbulent motions in the reactor are important, these motions
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46 being strongly linked with the bubbles dispersion and other gas-liquid phenomena [14].
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50 From a chemical engineering viewpoint, the foam is considered as a special case of gas-liquid
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52 dispersion with a large air/liquid ratio [15]. The foam formation in a sparged bioreactor is a
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54 dynamic phenomenon involving two competing processes, i.e. the rate of disintegration of
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56 bubbles in the foam and the rate of bubbles entering the foam. Freshly formed foam is
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58 composed of spherical bubbles separated by liquid films having up to a few millimeters in
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thickness. The foam stabilization occurs if the bubbles remains separated from each other by a liquid lamella, a mechanism favored by the presence of surface active molecules. During the ascension of the foam column in the headspace of the bioreactor, the liquid contained in the lamella is drained. At this stage, the aged foam is morphologically different of the fresh foam as it contains polyhedral in which three bubbles intersect at a corner with an angle of 120° [15] (these corners are called Plateau border). The drainage of the liquid of this lamella is responsible of the collapse of the foam. Like the other gas-liquid dispersion, the destabilization of the foam involves the aggregation of the gas bubbles known as the coalescence phenomenon. The evolution of gas bubbles from the bulk of the reactor to the headspace, as well as the shape modification of the air bubbles entrapped in the foam phase is depicted at figure 3. As shown in this figure, the generation of the foam depends on the rate of alimentation in air bubbles through the liquid surface. The foam rise velocity in the headspace of the reactor is thus strongly dependent of the gas rise velocity in the bulk of the reactor.

$$V_{foam} = f(V_g) = f\left(\frac{G}{S}\right) \quad (2)$$

The gas superficial velocity is described as the ratio of the air flow rate G (m³/s) on the reactor cross-sectional area S (m²).

In order to limit the foam formation in bioreactor, i.e. the gas-liquid dispersion at low liquid fraction occurring in the headspace of the reactor, it is thus important to understand the gas-liquid dispersion at high liquid fraction occurring in the bulk of the reactor. This is a strong problem owing to the incompatible nature of the two dispersions. Indeed, the gas-liquid dispersion is favored in order to maximize oxygen transfer, but in turn, this dispersion induces the appearance of foam. These two phenomena will be discussed in the following lines by highlighting the interactions between the oxygen transfer efficiency and the foam intensity. Air is injected into the bioreactor in order to ensure a continuous oxygen supply, oxygen

playing the role of the final electron acceptor in the microorganisms involved in aerobic processes. The oxygen transfer in bioreactor is described by the general equation:

$$\frac{dC_L}{dt} = k_L a (C_L^0 - C_L) \quad (3)$$

With $k_L a$ (s^{-1}), defined as the volumetric rate of oxygen transfer, being the main performance parameter used to estimate the gas-liquid transfer efficiency, C_L^0 being the dissolved oxygen concentration at saturation (typically around 8 mg/L) and C_L being the actual dissolved oxygen concentration. The bioreactor gas-liquid hydrodynamics is optimized in order to minimize the bubble size and to increase the specific surface available to transfer oxygen (represented by a in the $k_L a$). In the literature, two kinds of correlations putting the $k_L a$ in function of the operating parameters of the bioreactor have been proposed [16].

For bubble column or airlift bioreactors:

$$k_L a = C \cdot \left(\frac{G}{S} \right)^\alpha \quad (4)$$

For stirred tank reactors:

$$k_L a = C \cdot \left(\frac{P}{V} \right)^\alpha \left(\frac{G}{S} \right)^\beta \quad (5)$$

With G being the air flow rate (m^3/s), S being the reactor cross sectional area (m^2), P being the power drawn at the level of the impeller (W) and V being the volume of the reactor (m^3).

The gas-liquid dispersion in relation with $k_L a$ being far more described in the literature than the foaminess, and owing to the strong relation existing between the two kinds of dispersion [17], the operational parameters influencing oxygen transfer can be used to discuss about the foam influencing parameters at the process scale. In general, the critical condition for foaming in a bioreactor (with or without stirrer) is reached when the gas velocity exceeds a critical value depending on the liquid velocity in the bulk of the reactor [14]. The ratio between these two velocities can be modified by the operational parameters. In pneumatic reactor (e.g., bubble column and airlift reactors), the gas-liquid dispersion is quite simple to describe,

because only the air flow rate can be considered in order to improve the oxygen transfer (Equation 4). It has been shown that the foam appears in a bubble column bioreactor when the gas superficial velocity V_g is larger than a critical value for a fixed liquid velocity V_l [14]. The appearance of foam in such bioreactor can thus be described by a simple linear relationship :

$$V_g = a \cdot V_l + b \quad (6)$$

With a and b depending on the nature of the foaming solution.

From this equation, the foam increases when V_g increases for a fixed V_l , the foam decreases when V_g increases for a fixed V_l and no foam is produced when V_l is large enough. Similar results have been obtained in the case of 10 m³ jet loop bioreactor with a draft tube for which a stronger liquid circulation is correlated to a reduction of the foaminess of the medium [18].

The situation is more complicated for stirred bioreactor, for which the gas-liquid dispersion is governed by two operational parameters: the stirrer speed and the air flow rate (Equation 5). These two parameters must be considered together in order to describe the quality of the gas-liquid dispersion [19]. If the air flow rate is too important and/or if the stirrer speed is insufficient, the reactor is in the flooding regime. The air bubbles coming from the sparger cross the impeller cross-sectional area without being dispersed, leading to a weak specific area (a) and thus a low $k_L a$. The dispersion can be optimized by increasing the stirrer speed and/or by lowering the air flow rate. At this stage, the reactor is in the loading regime and the bubbles are mechanically broken when passing in the vicinity of the impeller leading to an effective gas-liquid dispersion. Our personal findings suggest that there is a direct relationship between the foam rise velocity in the headspace of the reactor and the gas-liquid dispersion regime in stirred bioreactor (Figure 4). These considerations suggest that the operational parameters of the bioreactors can be optimized in order to reach an acceptable level of oxygen transfer and to limit to some extent the foaminess. It has led, for example to the "Stirring As Foam Disruption" (SAFD) concept [20] that will be discussed in details further in the text.

However, in case of intensive foaming, a foam control procedure has to be considered (as it will be discussed in sections 6 and 7).

4. Methods used to evaluate the foaming properties of a solution

In order to estimate the problems associated with the formation of foam, it is important to have at disposition a technique for measuring some parameters linked with the foaming intensity. The foaming properties of proteins or surfactants solutions can be classified in two main categories: the foaming intensity (foaminess or foamability) and the foam stability. These two properties seem to be complementary, but can diverge in some conditions. For instance, a temperature increase leads to an increase of the foaminess and to a decrease of the foam stability [21]. In a general manner, the foamability of a surfactant solution refers to its overall capacity to produce foam, irrespective of some given characteristic of this foam (e.g., its liquid content, its bubble size distribution,...). On the other hand, the foam stability is a parameter taking into account the rate of change of the foam properties [13]. It is thus important to consider these two properties in order to characterize the foaming behavior of solutions.

Three main techniques have been used to determine the foaming properties of surfactants solutions. The first one is based on the volume of foam generated in function of time, known as the Bikerman method [22]. The foam can be generated by whipping or bubbling, but considering the lack of reproducibility of the former, the bubbling apparatus is preferred [23]. This apparatus comprises a glass column equipped with a porous sparger in order to disperse uniformly the air bubbles at the base of the column. The second method is based on the measurement of the liquid drainage from the foam. This method can be used only to measure the stability of the foam and not the foaming intensity of a media. The third one is based on

the conductivity of the foam, foam with a high water content exhibiting a higher conductivity. A more sophisticated method involves the combination of both the volume and the conductivity measurement, the two techniques being known to produce complementary results [5, 23]. The conductivity method is the most sophisticated and permits the measurement of the conductivity in multiple point of the foam column, allowing for example to have a precise description of the foam behavior in the headspace of a bioreactor [24]. This principle is currently under investigation in order to elaborate probe able to measure on line the development of foam inside a reactor (more details will be given in the next section). This kind of probe is based not only on the foaminess (related to the height of foam inside the reactor), but also to the foam stability, giving a clearer picture of what can be done to destroy the foam.

These techniques can also be used to test the efficiency of chemical antifoam [25]. In this case, foam is generated and a given quantities of antifoam is added when it reaches the top of the column. The frequency of antifoam addition on a given time interval can be exploited in order to measure the efficiency of the antifoam, the frequency of addition being inversely proportional to this efficiency.

5. Detection and regulation of foam in bioprocesses

The detection of foam is generally performed by using a resistivity probe located in the headspace of the bioreactor. The foam level is then detected by a contact probe inducing the automatic delivery of the antifoam. The contact between the probe and the foam close the electrical circuit supplying the antifoam delivery pump. However, this simple on/off control do not take into account the dynamics of foam formation and can leads to the delivery of an inappropriate dose of antifoam. Current researches involve more sophisticated probes able to

measure the conductivity at different levels of the foam column. The conductivity value can be exploited to measure the liquid content of the foam at a given point [24]. This method is notably used to estimate the foaming properties of solutions, as explained in section 4. This measure is more representative of the properties of the foam and notably its stability, giving a robust parameter that can be exploited to improve the dynamic control of foam during a bioprocess. However, such kind of sensor is sensitive to fouling by the residual foam. This problem can be avoided by an appropriate processing of the signal at the level of the transmitter. By this way, some companies (e.g. Charis) propose a sensor technology able to detect the foam intensity and to deliver in turn a precise dose of antifoam. As an example, the combination of the multi conductivity probe with a neural network processing of the signal has allowed an effective control of the foam during a *Pseudomonas sp.* cultivation performed in a laboratory scale stirred bioreactor [26].

The automatic control when using a physical defoamer can be based on the power input variation required by the turbine of the foam breaker when being exposed to foam [15].

6. Chemical methods used to prevent foam

An agent that inhibits initial foam formation is often called “antifoam”. An agent that ruptures already formed foam is often called a “defoamer” [27]. We will however use the word “antifoam” in a general way, whether it is used as antifoam or defoamer. Typical antifoams or defoamer consists of oil, hydrophobic solid particles or a mixture of both [28-31]. Non polar oil and polar oils have been described while solid particles can be an organic or inorganic. Non-polar oils are based either on silicone oils (polydimethylsiloxane, see vide infra) or organic oils. Most important organic non-polar oils are water insoluble paraffinic mineral oils, vegetable oils such as tall oil, soybean oil or rape seed oils or polymeric oil such

as polyisobutylene. Although some non-polar oils may exhibit some antifoaming performance on their own, mixtures of non-polar oil with hydrophobic solid particles perform much better than each components, which can be ineffective when used alone in some cases. Hydrophobic solid particles which are the most widely used in the making of antifoam formulations consist in treated silica (either fumed or precipitated silica), organic waxes (such as ethylene bis-stearamide) or so-called metallic soap (aluminum or Calcium stearate) [32]. There is a whole class of soluble foam control agents, mostly based on polypropylene (PO) or polyethylene-oxyde (EO) (or copolymers such as EO-PO-EO or ethoxylated fatty alcohols) which function by phase separating above their cloud point to form insoluble droplets of a hydrophobic liquid [33]. Some soluble polymeric surfactants are thought to destabilize foam stabilized by proteins or macromolecules by entering the surface as a soluble surface-active species and decreasing the surface elasticity. Antifoam with a surfactant structures are using this principles [34].

6.1. Modes of action of a chemical defoamer

Many foam control agents are added to the foaming medium after being predispersed in water, either as self-dispersible neat materials, oil-in-water emulsions, or self-dispersible mixtures or compounds. The main reason justifying this kind of formulation comes from the fact that the antifoam must be dispersed as small droplets in the liquid medium to be efficient. Indeed, in order to destroy a foam film, it is critical that a hydrophobic particle droplet emerges from the aqueous phase into the air/water interface (this process is termed as entering). Subsequently to the entering process, the oil coming from the inserted droplet is able to spread on the interface. Two coefficients, i.e. the entering and the spreading coefficient, are thus used to characterize the change in free energy of the system after the

addition of antifoam droplets. The entering coefficient can be described by the following equation:

$$C_{\text{entering}} = \sigma_{A/W} + \sigma_{O/W} + \sigma_{O/A} \quad (7)$$

A positive value for this coefficient means that the surface tension of the antifoam liquid ($\sigma_{O/A}$) is lower than the sum of the surface tension of the foaming solution ($\sigma_{A/W}$) and the interfacial tension between the antifoam and the foaming solution ($\sigma_{O/W}$). When an oil drop is spread over an air/water interface, the surface tension variation can be measured by the spreading coefficient given the following equation:

$$C_{\text{spreading}} = \sigma_{A/W} - \sigma_{O/W} - \sigma_{O/A} \quad (8)$$

During this step, the water surface is progressively replaced by the oil. A positive value for the spreading coefficient means that the surface tension of the foaming liquid is ($\sigma_{O/W}$) greater than the sum of the surface tension of the antifoam ($\sigma_{O/A}$) and the interfacial tension between the antifoam and the foaming liquid ($\sigma_{O/W}$). Entering is obviously essential to rupture the foam, and it is generally recognized that the entering coefficient must be positive to cause this rupture. A recent study suggests that the spreading of a layer of oils facilitate the foam breaking mechanism, suggesting that the ability to spread is an important property for an oil used in an antifoam formulation [35]. In order to get a positive value for both the entering and the spreading coefficient, it is important to have a liquid with a low surface tension, condition which is met for example in the case of silicone oils. It has been shown that hydrophobic particles facilitate the penetration of the antifoam droplets in the foam films, explaining the benefit associated with their addition in antifoam formulations [36]. On the other hand, the dispersion of hydrophobic particles in a reactor is an efficient method to control foam [37]. After a silicone droplet has penetrated the double air-water interface, a bridge is formed between the two surfaces of the foam film (Figure 5). At this stage, one of the mechanisms invoked in the literature involves the dewetting of the solution away from the

antifoam droplet (associated with the low surface tension of the oil) and leading eventually to the rupture of the film. By no mean, this is the only mechanism proposed in the literature but it helps to better understand how antifoam droplets can eventually lead to foam rupture.

6.2. Polydimethylsiloxanes as surface-active ingredients in antifoams

Silicone oils, and more specifically polydimethylsiloxane (PDMS) materials, combine many unusual properties because of their molecular characteristics, such as the flexibility of the Silicium-Oxygen-Silicium backbone and the very low cohesive energy existing between methyl groups (Figure 6). PDMS polymers have low surface tension, and most of them are nonvolatile and remain liquid even at quite high molecular weights. They are also highly insoluble in water [38]. Because of the extreme flexibility of the siloxane backbone and ease with which various polymer configurations can be adopted, and despite the siloxane backbone's considerable polarity, it is the polymer side groups that are the primary surface-active entities in the polymer structure. The pendant groups in PDMS are methyl groups, which show the weakest intermolecular interactions known: the London dispersion forces. The low surface tension, which is a direct manifestation of low intermolecular forces, confirms that the interactions between two PDMS chains occur only through their methyl groups. The polymer backbone controls the organization of the side groups at the surface, and its flexibility has a major effect on the ease with which the pendant groups can adopt preferred configurations. Thus, from a surface tension standpoint, the more flexible the backbone, the more readily will the lowest surface energy configurations be adopted. PDMS is a particularly favored case of very low intermolecular force pendant groups anchored along the most flexible backbone, thus allowing the methyl groups to be ideally presented to the external world. It is often observed that neat silicone oil shows low efficiency as antifoaming

agent. But mixtures of such oils with hydrophobic particles such as treated silica or finely divided high melting point waxes are generally much more effective than the individual components [28, 29, 36, 39]. In fact, the mixture performs well even if each component is ineffective when used alone. This synergy is observed for most combinations of oils and solids and in various types of foaming media. Effective foam control agents continue to be developed using such combinations to adjust for different types of foam problems.

6.3. Formulation trends

The delivery vehicles, by which the antifoam is delivered to the foaming system is a key aspect of the antifoam formulation. Practical aspects such as the physical state of the foam control agent (liquid, pasty or solid), its viscosity (which can be a limit for the pumping or dosing systems), the required cost efficiency, the addition method will all impact the way the foam control can or need to be added in the fermentation system. There are interaction between the parameters which control the foam formation (air incorporation, chemical composition of the broth – presence of surface active species or pH of the solution) and the antifoams characteristics (formulation of the antifoam) and how it is delivered to the foaming system. This is illustrated in figure 7.

6.3.1. Neat

Almost all foam control agent sold and used on the market are liquid. As discussed in a different section, they can belong to different chemical families (silicone, aliphatic oil, triglyceride, polyglycols, ethoxylated fatty alcohols, etc). The chemical composition will drive the physico-chemical properties, and amongst them the ease of dispersion into water. This is the main characteristic which will dictate if the foam control agent needs to be further

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3 formulated or can be used as such, which we will characterize as the neat active. If the foam
4 control agent has a low interfacial tension with water (or with the foaming solution), it may be
5 used as such as it may “disperse” in such a way it can reach the desired particle size and start
6 to control the foam. If the foam control agent show a high interfacial tension with water (like
7 polydimethylsiloxane or aliphatic oil with water), some further formulation will be required,
8 such as the addition of surfactant for the making of self-dispersible blend or as emulsion for
9 easier handling.
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19 20 21 22 6.3.2. Emulsions 23 24 25 26

27 Emulsions are disperse systems consisting of two (or more) mutually insoluble or sparingly
28 soluble liquids. The liquid usually present in excess is termed the continuous, or external
29 phase, while the liquid dispersed in it is termed the internal or dispersed phase. If the external
30 phase consists of water and the internal phase of an organic liquid, e.g., mineral oil, the term
31 “oil-in-water” (o/w) emulsion is used. In the field of foam control agent, emulsions are
32 generally aqueous, which means they are made by dispersing a non-aqueous foam control
33 agent (dispersed phase) into water (the continuous phase).
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43 Emulsions are prepared industrially to improve the effect of the foam control active substance
44 during the application by using it in a more or less finely dispersed form. The external phase
45 assumes the role of a transporting and/or diluting agent while the dispersed foam control
46 agent is present as tiny droplets.
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52 Emulsions are used for easy handling of foam control agents, which are sometime greasy,
53 highly viscous organic liquid. Stability of the emulsion is obtained thanks to the proper use of
54 surface active agents (surfactant) which reduce the interfacial tension between water and the
55 foam control agent, in such a way the high interfacial area between the dispersed phase and
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3 water can be achieved. Thickeners such as functionalised cellulose or some other natural or
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5 synthetic materials reduce creaming and further stabilise the emulsion. The latter can have
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7 some detrimental impact on some downstream treatment of the fermentation broth.
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10 11 12 6.3.3. Self-emulsifiable or self-dispersible systems 13 14 15

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17 Self-dispersible systems are blends of foam control agent with surfactants which are designed
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19 to readily disperse into the foaming system or readily emulsify and stabilize the foam control
20
21 agent in aqueous solutions prior to its addition in the foaming liquid. Various systems are
22
23 described in the literature such as dispersion of antifoam in silicone polyether, glycols or
24
25 organic surfactant. Although apparently simple, the compatibility between the foam control
26
27 agent and the surface active agent needs to be good enough in order to achieve stability of the
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29 self-dispersible systems, and yet effective enough to insure readily dispersion during the final
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31 application.
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38 39 6.3.4. Powdered antifoam 40 41 42

43 The last mean by which a foam control agent can be added is under a solid form. Most often,
44
45 this will be as a powdery material which can be conveniently dosed or handled. Various
46
47 methods exist to transform a liquid foam control agent into a solid powder. A simple
48
49 adsorption of a liquid on an absorbing solid is sometime enough to obtain a nice free flowing
50
51 powder. When improvement of the physical properties of the powder is required (i.e. free
52
53 flowing powder with improved shelf stability), more complex formulation and method of
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55 manufacturing will be used. This can imply the preliminary dispersion of the foam control
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57 active followed by spraying on a powder acting as carrier and water evaporation.
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7. Physical methods used to prevent or to reduce the foam formation in bioreactors

Various physical principles have been proposed in order to prevent foam formation [10]. It includes thermal method, ultrasound method and more classically the mechanical method involving a special turbine designed to break the foam bubbles when rotating. Considering that this latter method is by far the most widely used at the process scale, only this method will be investigated here (the ultrasonic method is actually under investigation but presents some adverse effects notably at the level of the influence of ultrasound on the viability of the microorganisms). In order to reduce the costs associated with the electrical motor, several turbines have been designed in order to be effective when placed directly on the shaft of the bioreactor. Finally, a new method, based on the management of the bioreactor hydrodynamics in stirred tank in order to reduce the foam formation, will be considered. This new method is termed as "Stirring As Foam Disruption" (SAFD). These three physical alternatives are presented at figure 8.

It is important to point out the fact that the first method proposed in this section can be used in order to suppress the foam, whereas the second and third method can only be involved in the limitation of its formation.

7.1. Mechanical foam breakers

7.1.1. Classical mechanical foam breaker

Considering the limitations associated with the chemical antifoams (see section 8), the use of a mechanical foam breaker can be in certain cases a good alternative. They can be used

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2
3 simultaneously with the addition of chemical antifoam in order to reduce the amount of
4 antifoam added in the bioreactor. There are numerous systems available which have been
5 extensively reviewed in the literature [10], and their actions are based on pressure change,
6 shear force, compressive force leading to the destabilization of the foam film. A good
7 example of commercially available and effective mechanical foam breaker is the Fundafom
8 system developed by Chemap (Figure 9) for which the turbines (inverted cones) have been
9 especially designed in order to enhance the shear effect. This apparatus is typically centrally
10 mounted on the bioreactor cover and allows an effective separation of the gas and liquid phase
11 (the volatile part of the collapsed liquid film is collected into a cooled condenser and injected
12 back into the reactor). Other system, including foam breaker with rotating disk [40], have also
13 been experimentally investigated. The main limitation of the mechanical systems is the energy
14 cost associated with the motor of the defoamer. In the case of a stirred vessel, two motor are
15 thus required, one for the mixing system, and another for the defoamer, leading to a
16 significant increase of the installation functioning and maintenance costs. These costs become
17 unacceptable in large-scale bioprocesses [1]. The following sections involve the description of
18 some costless alternatives to this system.

7.1.2. Mechanical foam breaker and mixing system mounted on the same shaft

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20
21 In order to reduce the costs associated with power consumption, the turbine of the mechanical
22 defoamer can be mounted directly on the shaft of the impeller system of the bioreactor. In
23 addition, this method presents less risks of contamination because the foam breaking
24 apparatus is directly integrated on the shaft of the bioreactor (no mechanical seals are
25 required). Considering the strong interrelation existing between oxygen transfer and foam
26 generation (as reported in section 3), this method is only valid if the critical speed for foam

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control is lower than the required speed for gas dispersion in the bulk of the bioreactor [41]. In this context, the turbine of the defoamer must be appropriately designed. A comparison has been performed between classical impellers (originally used for the mixing operation in the bulk of the reactor) and some newly designed turbine. It has been shown that these new turbines (thin needles and large paddle turbine) were more efficient in the sense that they lead to a significantly reduced critical speed for foam control [41-44]. A more recent system based on hollow inverted cones has been tested in pilot scale bioreactor [45]. This system is derived from the autonomous foam breaker Fundafom (see section 7.1.1.) and has allowed an efficient foam control during a *Bacillus* culture (a culture generally associated with strong foam problems considering the surfactant synthesis capabilities of this microorganism) at 0.5 m³ scale.

7.2. Stirring as foam disruption (SAFD) techniques

As discussed before, the bioreactor hydrodynamics can be optimized to limit foam formation and to ensure a given level of oxygen transfer. Further studies have led to the consideration of the hydrodynamics as a primary factor to prevent foam in stirred bioreactors, known as the "Stirring As Foam Disruption" (SAFD) concept. As seen before (section 3), the foaming intensity in relation with the gas-liquid dispersion in the reactor is governed by the ratio V_L/V_G . The SAFD technique consists simply to design the impeller system in order to enhance the liquid velocity near the free surface. In order to achieve this, the impeller configuration must be modified, i.e. the height of the upper impeller must be increased in order to be closer to the free surface. Several impeller geometries have been tested [20, 46, 47] and it has been shown that up-pumping hydrofoils (and notably the A340 hydrofoil commercialized by Lightnin) are the most effective because their flow pattern gives rise to

high liquid velocities in the vicinity of the free surface. Considering the complexity of the gas-liquid dynamics inside stirred bioreactors, the SAFD mechanism is not completely understood. In addition to the shear effects induced by the liquid flowing through the dispersion surface, the distortion of this surface under the action of liquid waves propelled by the hydrofoils can also play a role in the mechanism of foam rupture [48, 49].

8. Impact of foam formation and prevention on the bioprocess

The foam formation has several impacts at the level of a bioprocess design. In this section, the impact of foam formation will be reviewed by considering three distinct parts of the process : the cell growth and metabolites production, the bioreactor optimization and scale-up and the downstream processing operations. The appearance of foam induces a direct and an indirect negative effect on the bioprocess. The direct effect is attributed to the appearance of the foam which induces cell loss and sterility problems. The indirect effects are linked with the antifoam strategy. Mechanical foam breaker can break the cell membrane by shear effect. Chemical antifoam induces a lot of problems as they can alter the microbial metabolism, reduce the oxygen transfer efficiency and compromise the downstream processing operation.

8.1. Impact of foam formation and prevention on the microbial growth

The direct and indirect effects of foam on the bioprocess have to be distinguished. Foam affects directly the process by removing the cells from the medium by flotation. In addition, foam affects indirectly the process performances by the elaboration of a foam suppression strategy. Indeed, the addition of chemical antifoam has an impact on the microbial growth process. The non metabolizable antifoams, such as silicone oils and polyglycols, have little

effect on the microbial metabolism [11], although noticeable exception has been found in the case of exopolysaccharides production with the addition of a polypropylene glycol antifoam [50]. However, the process related problems associated with the use of such antifoam agents (i.e. principally oxygen transfer limitation and membrane fouling as it will be explained in the next sections) has led to the consideration of natural oils as an alternative. The use of vegetable oils have notably been investigated [51]. As the organic compounds (e.g. triglycerides and free fatty acids) contained in the oils can be used as a carbon source, the course of the fermentation is altered when using such product as antifoam. Their action on the metabolism depends on the microorganism cultivated and the composition of the oil: in given cases, they can increase the cell mass yield because they are incorporated as carbon source [51], and in other cases they can induce toxic effect related to the release of specific fatty acid [11]. A major disadvantage related to the use of natural oils as antifoam agent is their relative lower efficiency compared with their synthetic alternatives. In addition, the natural oil as to be added continuously during the culture as it is metabolized by the microorganism. Vardar-Sukan has made a comparison of the efficiency of several natural oils (castor, corn, cotton seed, linseed, olive, sesame, soybean and sunflower) and has found large variations at the level of their efficiency in function of the composition of the culture media, suggesting that the use of natural oil must be carefully tested before being applied to large-scale bioprocess [51].

8.2. Impact of foam formation and prevention on the bioreactor performances and scale-up

The use of chemical antifoam is known to reduce the oxygen transfer efficiency and in general the gas-liquid dispersion performances of the reactor. Indeed, the adsorption of

antifoam molecules at the gas-liquid interface induces a resistance to oxygen diffusion. In addition to this surface effect, the antifoam addition has a bulk effect interfering with the mixing mechanism and leading to reduced gas-liquid dispersion, larger bubbles and reduced gas hold up and interfacial area [52]. As seen before (section 2), many surface active agents are added as emulsions of suspended oil droplets in order to break the foam structure by formation of hydrophobic bridges and displacement of the proteins at the interface. However, these conditions induce in the same time the coalescence of the air bubbles in the bulk of the reactor [53]. These adverse effect are enhanced at higher aeration rate, leading to problems when the aeration process has to be scaled up [3]. This effect has been notably investigated and confirmed on 55L and 770L external loop airlift bioreactors with the addition of silicone based antifoams [53] and on bubble column bioreactors [54]. During the culture of plant cells, a significant decrease of k_{La} (see equation 2) has been observed in laboratory bubble column reactor and stirred tank reactor when little quantities of (as low as 20 ppm) polypropylene glycol and Pluronic PE6100 were added to the culture to control foam [55].

8.3. Impact of foam formation and prevention on the downstream processing operation

In general, the addition of antifoam during the culture leads to several problems at the level of the recovery procedures. The addition of antifoam modifies the suspension capabilities of the cells, leading to problems during the centrifugation and induces fouling of the micro and ultra filtration membranes [56, 57]. Considering their surface active properties, antifoams tend to stay at the interface, increasing their tendency to accumulate onto the membrane surface or pore walls. The antifoam polymers can easily spread on this surface, inducing the formation of a gel layer causing the fouling of the membrane. On a physico-chemical standpoint, several

mechanisms are involved in the membrane fouling by antifoams, e.g. hydrophobic interactions between emulsified antifoam and the membrane, cloud point effects of polyglycols and formation of a gel layer at the surface of the filtration membrane [58, 59]. The prevalence of one of this mechanism on the others depend on the nature of the filtration surface and the antifoam used. Pretreatment of polysulfone membrane surface has been proposed in order to reduce the fouling effect [60]. The treatment of the membrane with a wetting agent tends to decrease the hydrophobicity of the surface of the membrane. The hydrophylisation of the membrane can be achieved by chemical modification of the surface charge or by physical precoat of the membrane.

On the other hand, in some specific cases, the foam behavior of the broth can be exploited to perform a downstream processing operation. Indeed, protein of interest can be recovered by foam fractionation [61]. Specific methods for the direct purification of biosurfactants based on the foam generation from the culture media have also been proposed [62, 63]. On the same basis, the process of foam separation has been applied to separate microbial cells from the fermented medium in order to reduce the costs associated with centrifugation [64].

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List of figures

Figure 1: ideal system in which the surface tension σ decreases linearly with the surfactant concentration C .

Figure 2: difference of conformation of low molecular weight surfactant (left) and protein (right) at an air/water interface

Figure 3: macroscopic mechanisms involved in the generation and the aging of foam in an aerated bioreactor

Figure 4: relation between the foam rise velocity in a stirred bioreactor in function of the air flow rate and the stirrer speed. The transition between the flooding and the loading regime of the impeller has been highlighted. Experiments have been performed in a 300L stirred tank reactor equipped with a Rushton disk turbine with 6 blades.

Figure 5: schematic presentation of the bridging of a foam film by a spherical antifoam droplet. A : the antifoam droplet is entirely in the liquid film. B : the antifoam droplet has entered the surface. C : the antifoam droplet bridges the film. D : the process of bridge dewetting occurs, leading to destabilization of the foam film and eventually to foam rupture.

Figure 6: polydimethylsiloxane backbone of a silicon-based antifoam

Figure 7: scheme illustrating the different parameters taken into account in the elaboration of a chemical antifoam

Figure 8: different concepts of mechanical foam breaker. A : classical concept with the defoamer mounted on a separate shaft. B : defoamer with the turbine mounted on the shaft of the bioreactor. C : "Stirring As Foam Disruption" (SAFD) concept for which the liquid flow pattern generated by the upper impeller tends to destabilize the foam

Figure 9: classical mechanical foam breaker equipped with a rotating turbine and installed on a 20L bioreactor. 1 : motor ; 2 : shaft ; 3 : cooled condenser ; 4 : condenser outlet back to the reactor ; 5 : turbine (conical disks) ; 6 : cover part of the bioreactor

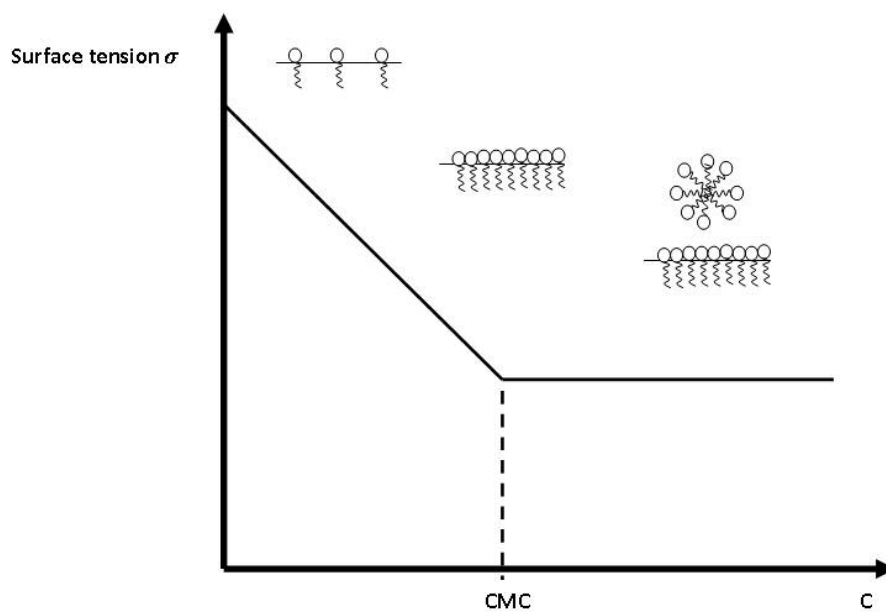
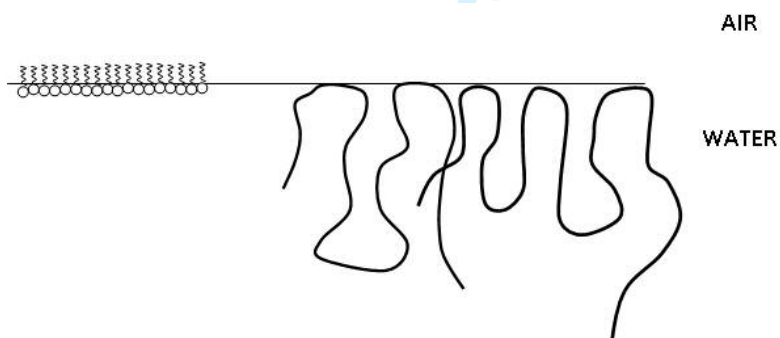
Figure 1**Figure 2**

Figure 3

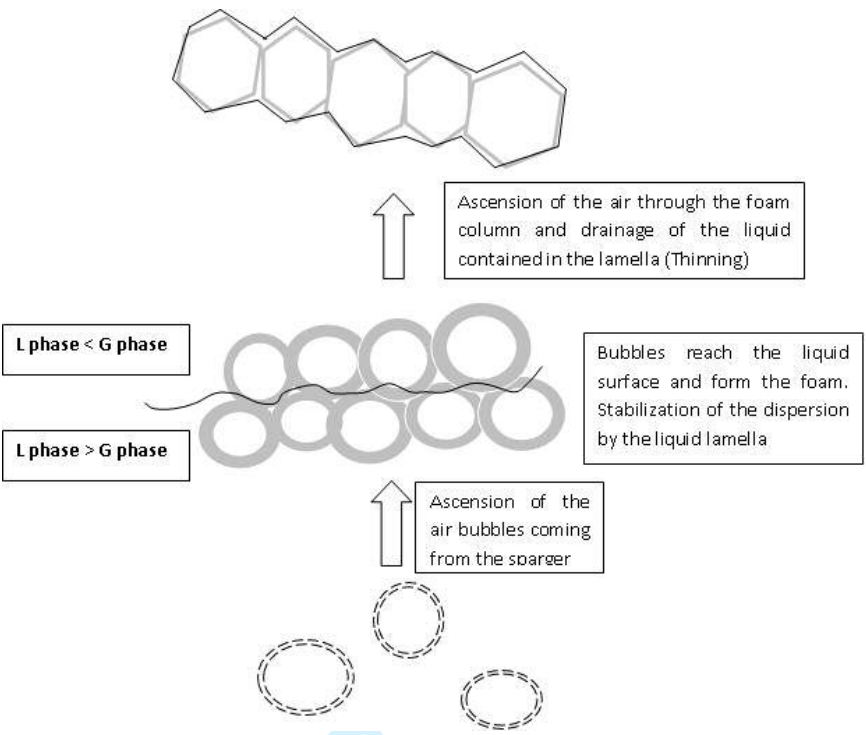


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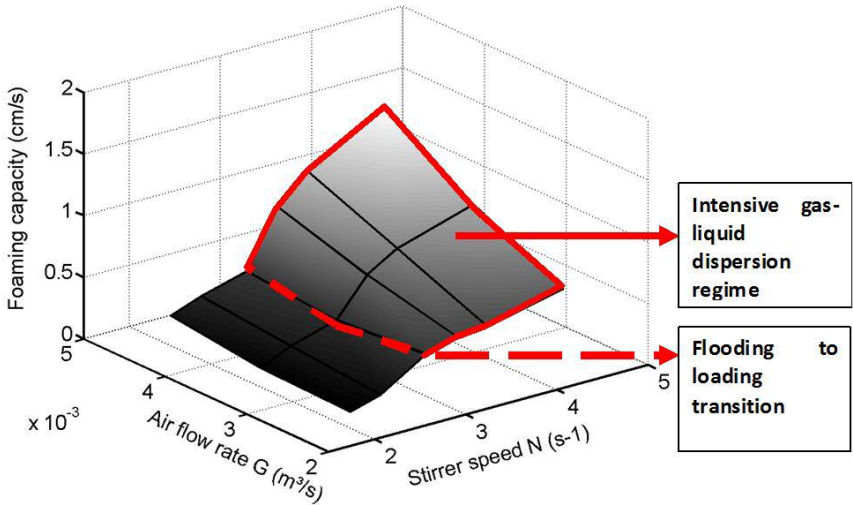


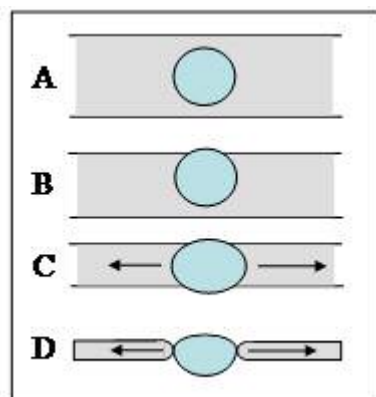
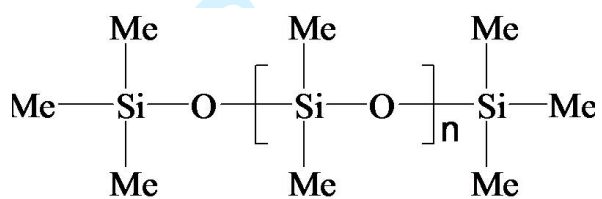
Figure 5**Figure 6**

Figure 7

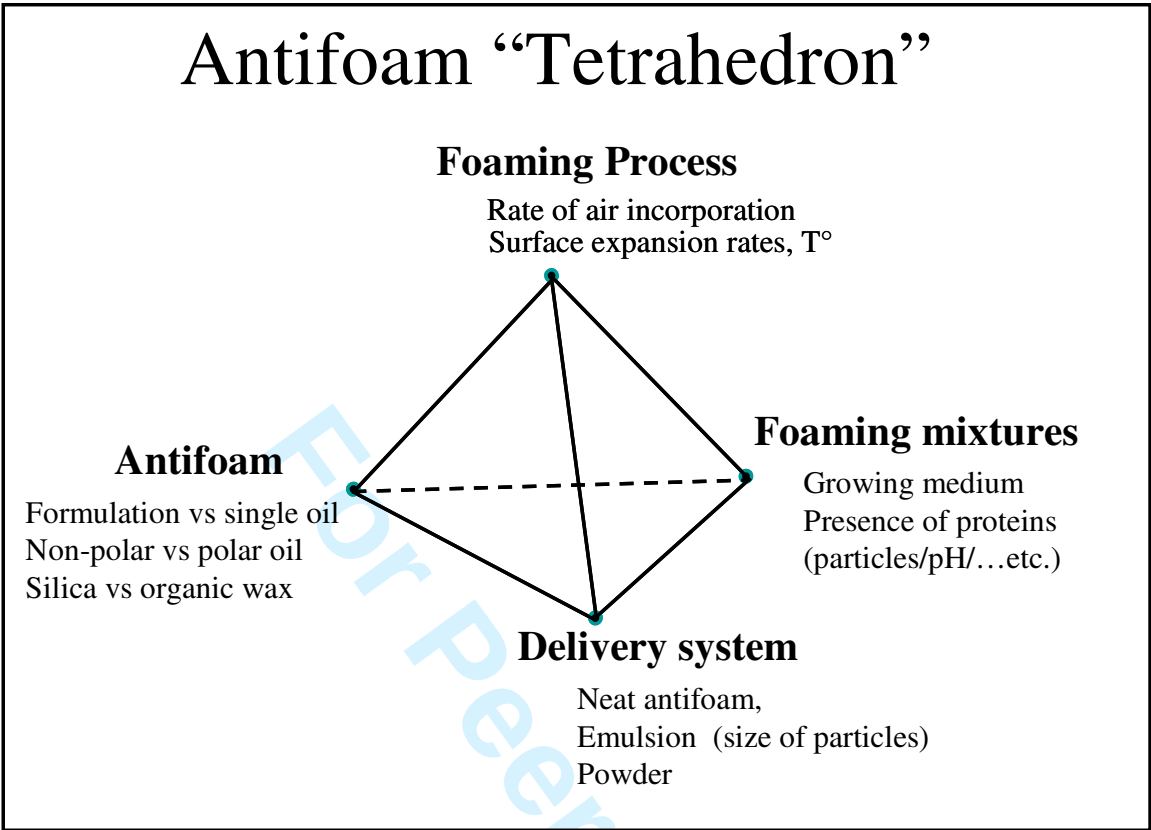


Figure 8

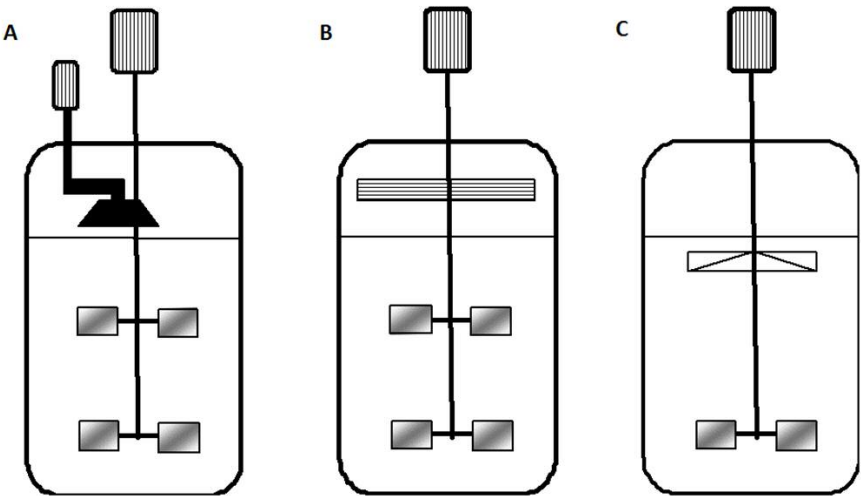


Figure 9

